

Sai Mulagan

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Summary

Data scientist with ~3 years of experience across the full ML lifecycle, from ETL and feature engineering to deployment and A/B testing. Strong in Python and SQL with production experience in LLMs, deep learning, and cloud infrastructure on AWS and GCP.

Education

Master's degree in Data Science & Artificial Intelligence, University of San Francisco August 2025 - June 2026

Bachelor's degree in Data Science, University of San Francisco August 2020 - May 2024

[AWS Certified Machine Learning Specialty](#) November 2024

[AWS Certified Cloud Practitioner](#) October 2024

Experience

Stanford University - Ophthalmic Informatics & AI Group ([OPTIMA](#)) Palo Alto, CA
Data Scientist September 2025 - Present

- Built scalable, reproducible ML pipelines in Python to train and evaluate deep learning models on 320K+ longitudinal patient records, improving progression prediction AUC by 12% and standardizing experimentation with MLflow across the research group.
- Fine-tuned LLMs and integrated RAG workflows with LangGraph and LangSmith to extract diagnoses, medications, and clinical measurements from 50K+ unstructured notes, achieving 80% extraction accuracy through structured evaluation and error analysis.
- Designed a multimodal data fusion framework aligning imaging features, tabular EHR data, and clinical text embeddings to forecast patient-level glaucoma progression, reducing visual field MD prediction RMSE by 18%.
- Collaborated with clinicians and AI researchers to evaluate explainability, bias, and robustness of LLM and computer vision models applied to medical imaging and clinical text.

Vertex Pharmaceuticals Boston, MA
Data Engineer, Intern May 2025 - August 2025

- Designed end to end ML pipelines and production APIs on Airflow and AWS SageMaker for medical data ingestion, feature engineering, and model training across 100K+ patient records, with tests and error handling, replacing 4 legacy workflows.
- Partnered with cross functional teams to translate business requirements into validated SQL datasets and Power BI reporting layers, documenting data definitions and usage guidance for stakeholders and reducing recurring reporting requests by ~30%.
- Automated ETL into Snowflake with data quality and validation checks across 5 sources, cutting data preparation time by ~40%.

Walmart Bentonville, AR
Data Scientist, Co-op May 2024 - May 2025

- Built XGBoost and LightGBM forecasting models across 500+ SKUs, reducing inventory misallocation by ~7% and saving an estimated \$1M annually across 5 regional distribution centers.
- Partnered with merchandising and supply chain teams to ship a price elasticity model covering 200+ product categories, informing markdown decisions that lifted gross margin by 3% on targeted SKUs.
- Built A/B testing framework for promotional offers, running 6 experiments that informed pricing strategy across 3 product categories.
- Developed a customer segmentation model using K-Means and RFM analysis on 5M+ shopper profiles, enabling 4 marketing campaigns that improved response rates by 15% over prior baselines.

Data Scientist, Intern May 2023 - August 2023

- Analyzed 10M+ transactions writing complex SQL (window functions, CTEs, multi-table joins) and Python, surfacing cross-sell opportunities that lifted targeted campaign conversion by 8% and added an estimated \$400K in revenue.
- Designed 12 interactive Tableau dashboards tracking real-time sales, pricing, and regional performance, used by cross-functional teams for weekly planning reviews.

Projects

[Yelp Recommender System: End to End Distributed ML Pipeline](#) February 2026 - March 2026

- Built a distributed recommendation system using Apache Airflow, MongoDB, and Google Cloud Storage to process Yelp review data
- Trained a Funk SVD collaborative filtering model and achieved an RMSE of ~1.33 for predicting user ratings and recommendations.
- Deployed a FastAPI service on Google Cloud Run with Docker, enabling API access to recommendations stored in MongoDB.

[LeaseLens: End to End Document AI Pipeline with OCR and LLM Extraction](#) May 2026 - June 2026

- Engineered swappable LLM and offline regex extraction behind one interface, pulling 15 structured fields from unstructured lease PDFs using Python and FastAPI.
- Added Tesseract OCR for scanned PDFs and a rule-based layer that flags policy exceptions in under a second per document.

Skills

Programming & Analysis: Python, R, SQL, Java, Bash, Pandas, NumPy, SciPy, Matplotlib, Seaborn, Git, Power BI, Tableau, Streamlit

Big Data & Distributed Systems: Apache Airflow, Spark, PySpark, SparkSQL, Kafka, Snowflake, ETL Pipelines, MongoDB, NoSQL

Machine Learning: Scikit-learn, TensorFlow, PyTorch, XGBoost, Transformers, LLMs, NLP, SHAP, LIME, Recommender Systems

Cloud MLOps: AWS (S3, EC2, SageMaker, Glue), GCP, Docker, Kubernetes, Terraform, MLflow, FastAPI, Flask, CI/CD, A/B Testing